

NORTH BATTLEFORD, SK, Canada

WMO: 718760

Lat: 52.7691N

Lon: 108.2510W

Elev: 548

StdP: 94.91

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.4	-37.9	0.1	-33.1	-34.8	0.1	-30.2	12.8	-10.7	11.3	-10.1	2.1	300	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 29.3	(d) 18.0	(e) 27.4	(f) 17.2	(g) 25.7	(h) 16.6	(i) 19.9	(j) 26.5	(k) 18.7	(l) 25.0	(m) 17.7	(n) 23.7	(o) 5.2	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.4	23.0	16.4	12.5	21.7	15.3	11.6	20.5	59.5	26.6	55.6	25.1	52.0	23.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 10.0	(c) 9.0	(e) -36.8	(f) 33.3	(g) 3.2	(h) 2.5	(i) -39.1	(j) 35.1	(k) -41.0	(l) 36.6	(m) -42.8	(n) 38.0	(o) -45.2	(p) 39.8		
(a) 11.3	(b) 10.0	(c) 9.0	(e) -36.9	(f) 22.2	(g) 3.2	(h) 1.3	(i) -39.2	(j) 23.1	(k) -41.1	(l) 23.9	(m) -42.9	(n) 24.6	(o) -45.2	(p) 25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-14.9	-13.2	-6.5	3.5	10.7	15.3	17.8	16.7	11.7	3.8	-5.7	-13.0
	DBStd	13.33	8.82	7.72	7.80	5.84	4.60	3.40	2.94	3.46	4.55	4.93	6.74	8.09
	HDD10.0	3611	773	651	513	206	47	3	0	1	33	199	471	713
	HDD18.3	5935	1032	884	771	446	240	101	44	72	203	450	721	972
	CDD10.0	784	0	0	0	9	68	162	243	209	84	7	0	0
	CDD18.3	68	0	0	0	0	3	10	29	23	3	0	0	0
	CDH23.3	1014	0	0	0	5	73	160	352	323	98	2	0	0
	CDH26.7	248	0	0	0	1	14	34	83	92	24	0	0	0
Wind	WSAvg	4.0	3.8	3.8	4.4	4.8	4.6	4.1	3.7	3.7	4.0	4.1	3.9	3.7
Precipitation	PrecAvg	391	20	14	19	23	39	67	69	52	32	16	18	21
	PrecMax	496	40	39	39	82	137	186	145	126	84	46	43	38
	PrecMin	286	6	2	4	1	8	7	12	7	3	2	2	6
	PrecStd	54	10	10	11	17	24	41	30	37	23	11	12	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	4.7	12.4	23.0	28.5	30.0	31.4	32.5	29.6	22.3	11.9	6.2
		MCWB	2.1	1.7	6.4	11.4	14.7	17.5	20.0	18.4	16.6	12.5	6.7	2.9
	2%	DB	2.9	2.1	7.7	19.6	25.4	27.1	28.9	29.2	26.2	17.6	8.0	3.3
		MCWB	0.7	0.0	3.7	9.9	13.5	16.3	19.2	18.1	15.2	10.0	4.2	0.8
	5%	DB	0.5	-0.2	5.2	16.3	22.8	25.0	26.9	26.7	23.3	14.4	5.1	0.6
		MCWB	-1.1	-1.5	2.3	8.0	12.4	15.3	18.4	17.6	14.0	8.3	2.5	-1.0
	10%	DB	-2.6	-2.3	3.1	13.2	20.3	23.1	25.2	24.8	20.5	11.7	2.9	-2.0
		MCWB	-3.6	-3.3	1.0	6.4	11.2	14.6	17.7	16.6	12.7	7.0	0.9	-3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	1.9	6.6	12.4	16.5	19.6	22.2	20.7	17.6	13.2	6.8	3.2
		MCDB	4.6	4.5	11.5	21.8	25.5	26.3	28.7	27.5	26.3	20.8	11.0	5.6
	2%	WB	0.8	0.1	4.0	10.3	14.3	17.8	20.5	19.4	15.8	10.7	4.7	1.0
		MCDB	2.8	1.8	7.3	18.1	23.1	24.2	26.9	26.6	24.9	15.9	7.7	3.1
	5%	WB	-1.2	-1.5	2.4	8.6	13.0	16.6	19.3	18.3	14.5	8.9	2.6	-1.0
		MCDB	0.5	-0.2	4.9	15.5	21.4	22.7	25.2	25.0	22.1	13.7	4.8	0.7
	10%	WB	-3.6	-3.3	1.1	7.0	11.7	15.6	18.2	17.2	13.2	7.3	1.0	-3.2
		MCDB	-2.6	-2.3	3.0	12.6	19.1	21.5	23.9	23.5	19.4	11.3	2.7	-1.9
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	9.7	11.9	13.7	12.1	12.3	13.1	13.1	11.0	8.8	8.8
		MCDBR	11.0	11.5	11.5	17.2	17.5	16.0	15.4	16.9	18.4	16.2	11.9	11.5
		MCWBR	9.4	10.0	8.3	9.5	8.5	7.5	7.6	7.9	9.0	9.7	8.7	9.7
		MCDDBR	10.8	11.0	11.0	16.3	15.5	13.1	13.6	14.6	17.0	14.9	11.2	10.9
	5% WB	MCWBR	9.3	9.7	8.1	9.4	7.9	6.9	7.4	7.4	8.7	9.3	8.4	9.2
Clear-Sky Solar Irradiance	taub	0.257	0.252	0.291	0.341	0.360	0.375	0.374	0.378	0.318	0.279	0.255	0.224	
	taud	2.266	2.332	2.362	2.338	2.335	2.331	2.347	2.323	2.472	2.553	2.452	2.364	
	Ebn at Noon	742	881	906	893	890	876	870	844	867	845	763	743	
	Edh at Noon	61	78	95	111	117	119	116	112	85	63	51	46	
All-Sky Solar Radiation	RadAvg	1.05	2.12	3.61	4.75	5.83	5.94	6.20	5.03	3.60	2.10	1.03	0.76	
	RadStd	0.10	0.14	0.28	0.46	0.46	0.46	0.36	0.41	0.38	0.33	0.13	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page